Experiment 7: Bag of Words and TF-IDF Representation

A. Bag of Words (BoW) Model

```
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer

# Corpus from the lab manual
corpus = [
    "I loved the movie, it was it fantastic!",
    "The movie was okay, but not great.",
    "I hated the movie, it was terrible."
]

# Create and fit the vectorizer
vectorizer = CountVectorizer()
document_matrix = vectorizer.fit_transform(corpus)

# Get the vocabulary
feature_names = vectorizer.get_feature_names_out()

print("Vocabulary:")
print(feature_names.tolist())

# Display document-term matrix
print("\nDocument-Term Matrix:")
print(document_matrix.toarray())

# Display as a readable table
df_bow = pd.DataFrame(
    document_matrix.toarray(),
    columns=feature_names
)

print("\nWord Frequency Table:")
print(df_bow)

# Total frequency of each word across documents
word_frequencies = df_bow.sum(axis=0).to_dict()
print("\nWord Frequencies:")
print(word_frequencies)
```

```
Vocabulary:
['but', 'fantastic', 'great', 'hated', 'it', 'loved', 'movie', 'not', 'okay',

Document-Term Matrix:
[[0 1 0 0 2 1 1 0 0 0 1 1]
 [1 0 1 0 0 0 1 1 1 0 1 1]
 [0 0 0 1 1 0 1 0 0 1 1 1]]
```

```
Word Frequency Table:
   but  fantastic  great  hated  it  loved  movie  not  okay  terrible  the
0    0          1      0      0   2      1      1    0     0         0    1
1    1          0      1      0   0      0      1    1     1         0    1
2    0          0      0      1   1      0      1    0     0         1    1

   was
0    1
1    1
2    1

Word Frequencies:
{'but': 1, 'fantastic': 1, 'great': 1, 'hated': 1, 'it': 3, 'loved': 1, 'movi
```

B. TF-IDF Representation[link text]

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer

# Corpus from the lab manual
corpus = [
    "NLP is fun",
    "NLP helps computers understand language",
    "Language is power"
]

# Create TF-IDF vectors
tfidf_vectorizer = TfidfVectorizer()
tfidf_matrix = tfidf_vectorizer.fit_transform(corpus)

# Display feature names
features = tfidf_vectorizer.get_feature_names_out()

print("Vocabulary:")
print(features.tolist())

# Display the TF-IDF matrix
print("\nTF-IDF Matrix:")
print(tfidf_matrix.toarray().round(4))

# Display a labeled table
df_tfidf = pd.DataFrame(
    tfidf_matrix.toarray(),
    columns=features
)

print("\nTF-IDF Table:")
print(df_tfidf.round(4))
```

```
Vocabulary:
['computers', 'fun', 'helps', 'is', 'language', 'nlp', 'power', 'understand']

TF-IDF Matrix:
[[0.     0.6809 0.     0.5179 0.     0.5179 0.     0.    ]
 [0.4905 0.     0.4905 0.     0.373  0.373  0.     0.4905]
 [0.     0.     0.     0.5179 0.5179 0.     0.6809 0.    ]]

TF-IDF Table:
   computers     fun   helps      is  language     nlp   power  understand
0     0.0000  0.6809  0.0000  0.5179    0.0000  0.5179  0.0000      0.0000
1     0.4905  0.0000  0.4905  0.0000    0.3730  0.3730  0.0000      0.4905
2     0.0000  0.0000  0.0000  0.5179    0.5179  0.0000  0.6809      0.0000
```